

PHYSICS COMPREHENSIVE MASTER TEST (18 QUESTIONS)

Complete Curated Exam Based on Sets 55/7/1, 55/7/2, 55/5/2, SR5QP/5

TIME: 3 HOURS

MAX MARKS: 70

SECTION A: OBJECTIVE & CONCEPTUAL

Q1. Two nuclei have mass numbers in the ratio 27:64. The ratio of their nuclear densities is:

Set 55/7/1, 55/5/2

- (A) 3 : 4 (B) 9 : 16
(C) 1 : 1 (D) 27 : 64

Q2. A p-n junction diode is forward biased. As a result, the potential barrier height:

Set 55/7/2, SR5QP/5

- (A) Increases (B) Decreases
(C) Becomes Zero (D) No change

Q3. Which part of the electromagnetic spectrum is used in water purifiers to kill germs?

Set 55/5/2

- (A) Infrared rays (B) Ultraviolet rays
(C) X-rays (D) Microwaves

Q4. The radius of the nth orbit in Bohr's model of the hydrogen atom is proportional to:

Set 55/7/1, 55/7/2

- (A) n (B) n^2
(C) $1/n$ (D) $1/n^2$

Q5. Plot of stopping potential (V_0) vs frequency (ν) for a photosensitive metal. The slope represents:

Set 55/7/1, 55/5/1

- (A) h (B) h/e
(C) e/h (D) Work function

SECTION B: VERY SHORT ANSWER (2 MARKS)

Q6. Draw energy band diagrams of n-type and p-type semiconductors at temperature $T > 0$ K. Show donor/acceptor levels.

Set 55/5/2, SR5QP/5

Q7. Two wires (same material) of length L & 2L and areas A & 2A are in series with a battery. Find drift velocity ratio.

Set 55/7/2

Q8. Calculate the Binding Energy per nucleon of a Helium nucleus (${}^4_2\text{He}$) given constituent masses.

Set 55/7/1

SECTION C: SHORT ANSWER (3 MARKS)

Q9. Define a wavefront. Using Huygens' Principle, verify the laws of Refraction (Snell's Law) at a plane surface.

Set 55/5/2, SR5QP/5

Q10. Using Bohr's postulates, derive the expression for the radius of the nth orbit of H-atom.

Set 55/7/1, 55/7/2

Q11. Plot Stopping Potential vs Frequency. (i) What does x-intercept show? (ii) Find work function from graph.

Set 55/7/1

Q12. Derive expression for force per unit length between two parallel current conductors and define 1 Ampere.

Set 55/5/2

Q13. Calculate electrostatic potential energy of a two-charge system in an external electric field $E = A/(r^2)$.

Set SR5QP/5

SECTION D: CASE STUDY (4 MARKS)

Q14. Case Study: Drift Velocity & Resistivity.

Set 55/7/1, 55/7/2

(i) Define Relaxation Time.

(ii) Relate Current density (J) and drift velocity (vd).

(iii) How does resistivity of semiconductor change with temperature?

Q15. Case Study: Lens Maker's Formula.

Set 55/5/2

(i) Double convex lens: $r_1=10\text{cm}$, $r_2=15\text{cm}$, $n=1.5$. Find focal length.

(ii) What happens to ' f ' if immersed in water?

SECTION E: LONG ANSWER (5 MARKS)

Q16. (a) Ray diagram of Astronomical Telescope (Normal Adjustment) & derive Magnification.

Set 55/7/1, 55/7/2, SR5QP/5

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(b) Ray diagram of Compound Microscope & derive magnification for image at near point.

Q17. (a) Working of p-n junction as Full Wave Rectifier with circuit and waveforms.

Set 55/7/2, 55/5/2

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(b) Principle and working of Moving Coil Galvanometer. Explain radial field & Voltmeter conversion.

Q18. (a) Derive Capacitance of Parallel Plate Capacitor with dielectric slab insertion.

Set SR5QP/5

--- OR ---

(b) Derive Electric Potential due to an electric dipole at a point on its axis.